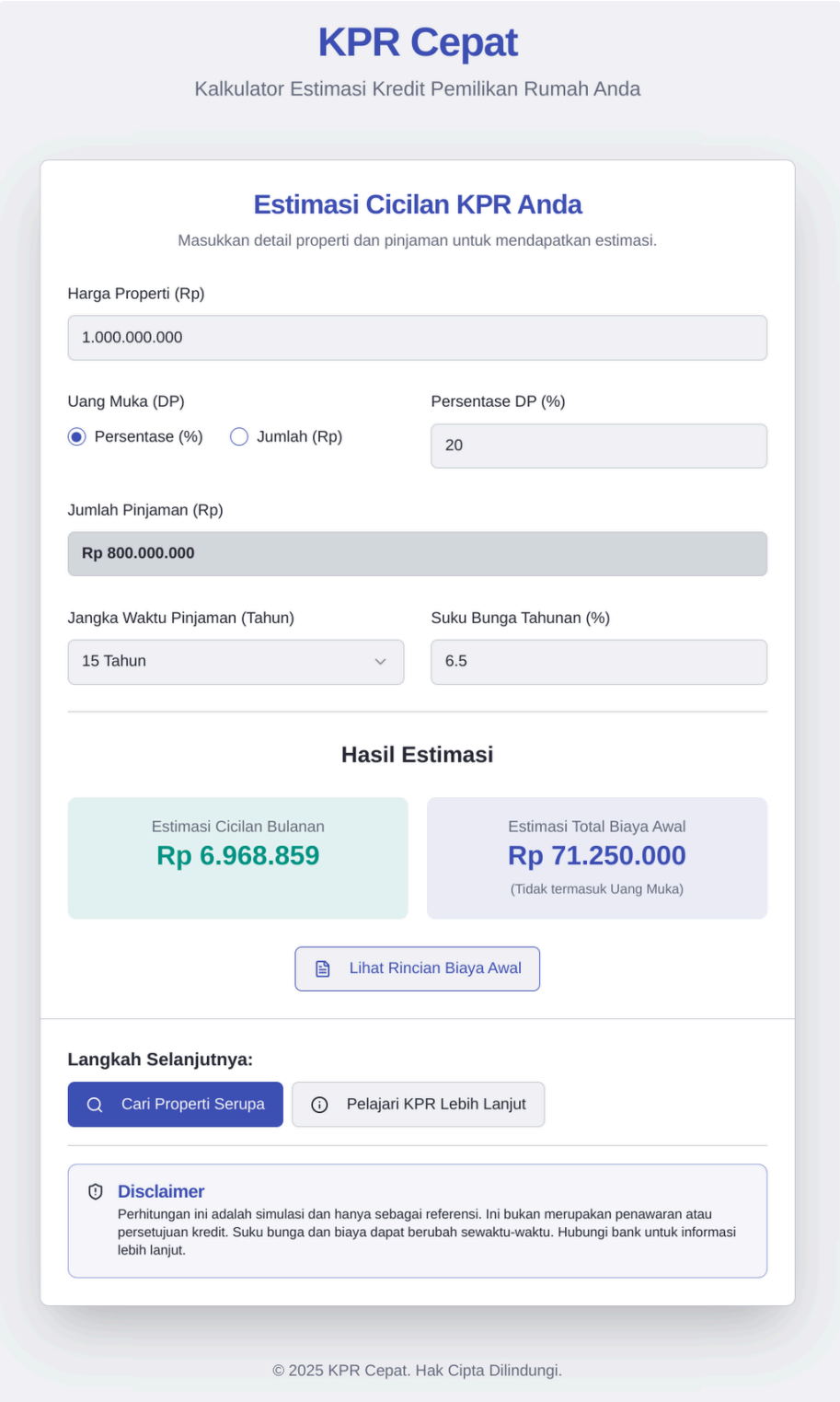
[PRD] Homepage Mortgage Estimator (KPR Estimator)

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1. Introduction

This document outlines the product requirements for a new Mortgage Estimator (KPR Estimator) feature to be integrated into the homepage of our property e-commerce platform. The primary goal of this feature is to provide prospective property buyers with a quick and easy way to estimate their potential monthly mortgage payments and associated upfront costs. This tool aims to enhance user engagement, provide valuable financial insights early in the property search journey, and potentially generate leads for partner banks or in-house mortgage services. The estimator will be tailored to the Indonesian market, using relevant terminology (e.g., KPR, Uang Muka, BPHTB) and considering common local fees and charges associated with property purchase financing.



2. Goals

- **Primary Goal:** Empower users to quickly estimate their potential monthly KPR (Kredit Pemilikan Rumah - Home Ownership Loan) installments and significant initial costs for properties they are interested in.
- **Secondary Goals:**
  - Increase user engagement on the homepage and overall platform.
  - Improve user experience by providing immediate, valuable financial planning information.
  - Educate users about the various costs involved in a property purchase beyond the down payment.

- Generate qualified leads for KPR applications (to be explored in future iterations).
- Position the platform as a comprehensive and supportive resource for the property buying journey in Indonesia.

3. Target Audience

- **Primary:** Prospective homebuyers in Indonesia actively browsing the e-commerce platform. This includes:
  - First-time homebuyers exploring affordability.
  - Existing homeowners considering upgrading or investing in additional properties.
  - Users in the early to mid-stages of their property search, seeking to understand financial implications.
- **Secondary:**
  - Property agents using the tool as a quick reference for their clients.
  - Internal sales and customer support teams assisting users.

4. Assumptions

- Users will have a basic understanding of property and loan concepts (e.g., property price, down payment/DP, loan term/tenor).
- The platform can maintain an indicative average KPR interest rate, which can be updated periodically.
- The initial version will focus solely on estimation and will not involve pre-qualification, credit scoring, or formal KPR application submission.
- The estimator will primarily address conventional KPR. Sharia-compliant financing (KPR Syariah) may be considered for future iterations.
- The tool will be prominently displayed or easily accessible on the homepage to maximize visibility and utility.

5. User Stories

- **US-001:** As a prospective homebuyer, I want to input the property price, my desired down payment (Uang Muka) percentage or amount, and preferred loan tenure (jangka waktu), so that I can instantly see an estimated monthly KPR installment (estimasi cicilan bulanan).
- **US-002:** As a user, I want to be able to input a specific annual interest rate (suku bunga tahunan) or use a pre-filled average market rate, so that the mortgage estimation is as realistic as possible for my scenario.
- **US-003:** As a user, I want to see a clear breakdown of the estimated upfront costs (estimasi biaya awal) associated with the KPR, such as BPHTB, notary fees, and bank administration fees, so I can understand the total initial financial commitment required.
- **US-004:** As a user, I want the mortgage estimator to be clearly visible and easily accessible directly from the homepage.
- **US-005:** As a user, I want the interface to be simple, intuitive, and presented in Bahasa Indonesia.
- **US-006:** As a user, I want to be able to adjust the input values (property price, down payment, tenure, interest rate) and see the estimated payments and costs update dynamically in real-time.
- **US-007:** As a user, after receiving the estimation, I want to see a clear Call to Action (CTA) suggesting relevant next steps, such as "View Properties in this Price Range" or "Learn More about KPR."
- **US-008:** As a platform administrator, I want to be able to easily update the default average interest rate and other configurable fee percentages/amounts used in the calculator through a backend interface.
- **US-009:** As a user, I want to clearly understand that the figures displayed are estimates for informational purposes only and may differ from actual bank offers.

- **US-010:** As a user, I want the input for down payment to be flexible, allowing me to enter either a percentage of the property price or a fixed Rupiah amount.
- **US-011:** As a user, I want the loan amount (jumlah pinjaman) to be automatically calculated and displayed based on the property price and down payment inputs.

6. Product Features

6.1. User Interface (UI) & User Experience (UX)

- **F-001: Homepage Estimator Form**
  - **Description:** A dedicated, visually distinct section or an easily launchable modal/widget on the homepage for the KPR Estimator.
  - **UI Elements (All in Bahasa Indonesia):**
    - **Judul (Title):** e.g., "Estimasi Cicilan KPR Anda" (Estimate Your KPR Installment)
    - **Input "Harga Properti (Rp):"** Numeric input for Property Price in Indonesian Rupiah.
    - **Input "Uang Muka (DP):"**
      - Option to input as "Persentase (%)" or "Jumlah (Rp)".
      - A toggle/radio button to switch between percentage and fixed amount input.
      - Numeric input field.
    - **Display "Jumlah Pinjaman (Rp):"** Auto-calculated (Property Price - Down Payment), read-only.
    - **Input "Jangka Waktu Pinjaman (Tahun):"** Dropdown or slider for Loan Tenure in years (e.g., common options: 5, 10, 15, 20, 25, 30 years).
    - **Input "Suku Bunga Tahunan (%):"** Numeric input for Annual Interest Rate. Pre-filled with a configurable average market rate but editable by the user.
  - **Interaction:** Results (Monthly Installment and Upfront Costs) update in real-time as user types or changes input values.
  - **Language:** Bahasa Indonesia.
  - **Currency:** IDR (Rp) displayed clearly.
- **F-002: Estimation Results Display**
  - **Description:** Clear, concise, and prominent display of the KPR estimation results.
  - **UI Elements (All in Bahasa Indonesia):**
    - **"Estimasi Cicilan Bulanan (Rp):"** Prominently displayed calculated monthly installment.
    - **"Estimasi Total Biaya Awal (Rp):"** Prominently displayed sum of all estimated upfront costs (excluding down payment, which is an input).
    - **Link/Button "Lihat Rincian Biaya Awal":** To expand or show a modal with the breakdown of upfront costs.
  - **Interaction:** Smooth transitions for displaying and updating results.
- **F-003: Detailed Upfront Cost Breakdown**
  - **Description:** A detailed, itemized list of estimated initial costs associated with acquiring the property and KPR. This can be shown in an expandable section or a modal.

- **UI Elements (Estimates, with configurable default percentages/amounts where applicable, all in Bahasa Indonesia):**
  - **"Uang Muka (DP):"** The amount entered/calculated.
  - **"Biaya Provisi Bank:"** e.g., 1% of Loan Amount. (Default percentage configurable).
  - **"Biaya Administrasi Bank:"** Fixed amount or percentage of loan. (Default configurable).
  - **"Biaya Appraisal:"** Fixed amount. (Default configurable).
  - **"Biaya Notaris & AJB (Akta Jual Beli):"** Typically 0.5% - 1% of Property Price. (Default percentage configurable).
  - **"BPHTB (Bea Perolehan Hak atas Tanah dan Bangunan):"** Calculated as 5% x (Harga Properti - NJOPTKP).
  - **NJOPTKP (Nilai Jual Objek Pajak Tidak Kena Pajak):** For V1, use a configurable, general average NJOPTKP for Indonesia (e.g., Rp 60,000,000 or Rp 80,000,000). Include a note that this value varies significantly by region.
  - **"Biaya Asuransi Jiwa (Perkiraan):"** Include a note: "Estimasi, tergantung usia & bank." (Estimate, depends on age & bank). For V1, this could be a very rough placeholder calculation (e.g., a small percentage of loan amount) or clearly marked as "Variatif" and not included in the sum, with a clear note.
  - **"Biaya Asuransi Kebakaran (Perkiraan):"** Include a note: "Estimasi, tergantung nilai properti & bank." (Estimate, depends on property value & bank). Similar to life insurance for V1.
  - **"Biaya Balik Nama (BBN) Sertifikat:"** Typically around 0.5% - 1% of Property Price (related to AJB/Notary, often bundled). (Default percentage configurable).
  - **Disclaimer for Costs:** A clear statement that "Semua biaya di atas adalah estimasi dan dapat berbeda tergantung pada bank, notaris, lokasi properti, dan kebijakan yang berlaku." (All above costs are estimates and can vary depending on the bank, notary, property location, and applicable policies.)
- **F-004: Call to Action (CTA) Buttons/Links**
  - **Description:** Strategically placed CTAs to guide users after they receive their estimation.
  - **UI Elements (Examples, in Bahasa Indonesia):**
    - "Cari Properti dengan Cicilan Serupa" (Links to a pre-filtered search results page based on the estimated monthly payment).
    - "Pelajari Lebih Lanjut tentang KPR" (Links to an informational page or blog articles about KPR in Indonesia).
    - (Future Iteration) "Simpan Estimasi Ini" or "Hubungi Konsultan KPR".
- **F-005: General Disclaimer**
  - **Description:** A mandatory disclaimer visible with the calculator results.
  - **UI Element:** Text such as: "Perhitungan ini adalah simulasi dan hanya sebagai referensi. Ini bukan merupakan penawaran atau persetujuan kredit. Suku bunga dan biaya dapat berubah sewaktu-waktu. Hubungi bank untuk informasi lebih lanjut." (This calculation is a simulation and for reference only. It is

not a loan offer or approval. Interest rates and fees may change at any time. Contact the bank for more information.)

6.2. Functional Requirements

- **F-006: Monthly Installment Calculation**
  - **Description:** Calculate the fixed monthly mortgage payment using the standard annuity formula.
  - **Formula:**
    $$ M = P \frac{r(1+r)^n}{(1+r)^n - 1} $$
    Where:
  - M = Estimasi Cicilan Bulanan (Monthly Installment)
  - P = Jumlah Pinjaman (Principal Loan Amount = Property Price - Down Payment)
  - r = Suku Bunga Bulanan (Monthly Interest Rate = Annual Interest Rate / 12 / 100)
  - n = Jumlah Pembayaran (Total Number of Payments = Loan Tenure in years * 12)
  - **Logic:** Ensure correct handling of DP (percentage vs. fixed amount) to derive P.
- **F-007: Total Upfront Costs Calculation**
  - **Description:** Calculate the sum of all applicable initial costs as detailed in F-003 (excluding the down payment itself, which is an input).
  - **Logic:** Sum the calculated values for Provisi, Administrasi, Appraisal, Notaris/AJB, BPHTB, and BBN based on their respective formulas/percentages of either loan amount or property price. Insurance costs might be excluded from the sum in V1 if too variable, but listed with disclaimers.
- **F-008: Input Validation**
  - **Description:** Implement client-side and server-side validation for all user inputs.
  - **Rules (Examples):**
    - Harga Properti: Must be a positive number. Minimum and maximum values may apply.
    - Uang Muka (%): Must be between 0% and 100%.
    - Uang Muka (Rp): Must be a positive number and less than or equal to Harga Properti.
    - Jangka Waktu Pinjaman: Must be a positive integer, selected from predefined valid tenures (e.g., 1 to 30 years).
    - Suku Bunga Tahunan: Must be a positive number, within a realistic range (e.g., 1% to 35%).
  - **Feedback:** Clear, user-friendly error messages in Bahasa Indonesia displayed next to the respective fields or in a summary.
- **F-009: Dynamic Recalculation**
  - **Description:** All calculated results (Monthly Installment, Loan Amount, Upfront Costs) must update automatically and immediately in the UI whenever any relevant input field value is changed by the user, without requiring a page reload.
- **F-010: Backend Configuration for Defaults**
  - **Description:** Provide an interface or configuration file for administrators to easily update:
    - Default average Annual KPR Interest Rate.
    - Default percentages or fixed amounts for all upfront cost components (Provisi, Administrasi, Appraisal, Notaris/AJB, BBN).

        - Default NJOPTKP value for BPHTB calculation.

6.3. Data Requirements (primarily for backend configuration)

- **D-001:** Default current average KPR annual interest rate in Indonesia.
- **D-002:** Default percentages/fixed amounts for KPR fees (provisi, admin, appraisal, notary/AJB, BBN).
- **D-003:** Default NJOPTKP value for Indonesia (or a simplified regional average).

7. Non-Functional Requirements

- **NF-001: Performance**
    - The calculator interface should load quickly as part of the homepage (e.g., LCP within 2.5 seconds).
    - Real-time calculations and UI updates should be perceived as instantaneous by the user (e.g., within 200-500ms of input change).
- **NF-002: Usability & Accessibility**
    - The form must be highly intuitive, requiring minimal instruction for users unfamiliar with financial calculators.
    - Fully responsive design, ensuring optimal user experience on desktop, tablet, and mobile devices. Test on common screen sizes in Indonesia.
- **NF-003: Reliability & Accuracy**
    - Mathematical calculations must be accurate as per the specified formulas.
    - The tool should be consistently available (high uptime).
- **NF-004: Security**
    - As no sensitive personal user data is collected or stored in V1 of the estimator, primary security focuses on protecting the integrity of the configuration data (interest rates, fees).
    - All data transmission to/from the server (if any for logging or future enhancements) must use HTTPS.
    - Standard protection against common web vulnerabilities (XSS, CSRF).
- **NF-005: Maintainability**
    - The codebase should be well-documented, modular, and easy to update (e.g., for changes in fee structures, interest rate logic, or UI).
    - Configuration of default values (interest rates, fees) should be straightforward for non-technical administrators.
- **NF-006: Localization & Internationalization**
    - **Primary Language:** Bahasa Indonesia. All UI text, error messages, and disclaimers must be in clear and natural Bahasa Indonesia.
    - **Currency:** Indonesian Rupiah (Rp), formatted correctly (e.g., Rp 1.000.000).

8. Release Criteria

- All User Stories defined for Version 1.0 are fully implemented and have passed QA testing.
- Successful completion of User Acceptance Testing (UAT) by the product team and key Indonesian stakeholders (e.g., local marketing or sales team).
- Performance criteria (NF-001) are met on targeted devices and network conditions.
- No critical or high-severity bugs identified in the KPR estimator functionality.

- All UI text, disclaimers, and informational content are reviewed and approved for accuracy and clarity in Bahasa Indonesia.
- Responsive design is verified across the most common browsers (Chrome, Firefox, Safari) and device types (desktop, popular Android models, iPhones) used in Indonesia.
- Backend configuration for default values is functional and tested.

9. Success Metrics

- **SM-001: Usage Rate:**
  - Number of unique users interacting with the KPR estimator per day/week/month.
  - Percentage of homepage visitors who engage with the estimator.
- **SM-002: Engagement:**
  - Average number of estimations (interactions with input fields leading to recalculation) per user session.
  - Click-Through Rate (CTR) on the CTAs provided (e.g., "Cari Properti," "Pelajari Lebih Lanjut").
- **SM-003: User Satisfaction (Qualitative):**
  - Feedback collected via on-site surveys (if implemented) or through general customer support channels.
  - Observed reduction in basic affordability questions to customer support.
- **SM-004: Task Completion Rate:**
  - Percentage of users who input all required fields and receive an estimation.
- **SM-005: Impact on Property Search (Exploratory):**
  - Analyze if users who use the calculator subsequently view more property details pages or perform more refined property searches.

This Product Requirements Document serves as the definitive guide for the development of the Homepage Mortgage Estimator (KPR Estimator). It will be subject to review and updates as the project progresses and more detailed information or feedback is obtained.